

Equation IDs: section.row (10.3 = Section 10, row 3). Linked equations on one row share an ID.

1 · Constants, charge, vectors

$$1.1 \quad k_e = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9, \quad \epsilon_0 = 8.85 \times 10^{-12}$$

$$1.2 \quad e = 1.602 \times 10^{-19} \text{ C}, \quad G = 6.674 \times 10^{-11}, \quad q = Ne$$

$$1.3 \quad Q_{\text{before}} = Q_{\text{after}}, \quad |\mathbf{E}| = \sqrt{E_x^2 + E_y^2}, \quad \theta = \text{atan2}(E_y, E_x)$$

k_e : Coulomb constant [$\text{N m}^2/\text{C}^2$]; ϵ_0 : vacuum permittivity [F/m]; G : gravity constant [$\text{N m}^2/\text{kg}^2$]; e : elementary charge; N : signed integer. $\mathbf{E}$: field [N/C = V/m]. $\mu = 10^{-6}$, $n = 10^{-9}$, $p = 10^{-12}$.

An isolated system conserves net charge. Gaining electrons makes an object more negative. Bold symbols are vectors; hats are unit vectors. Angles in calculus are radians.

2 · Coulomb force, gravity & superposition

$$2.1 \quad \mathbf{R}_i = \mathbf{r} - \mathbf{r}_i, \quad \mathbf{E}(\mathbf{r}) = k_e \sum_i \frac{q_i \mathbf{R}_i}{|\mathbf{R}_i|^3}, \quad \mathbf{F} = q_0 \mathbf{E}$$

$$2.2 \quad F_C = k_e \frac{|q_1 q_2|}{r^2}, \quad F_g = G \frac{m_1 m_2}{r^2}, \quad U_g = -\frac{G m_1 m_2}{r}$$

$\mathbf{r}$, $\mathbf{r}_i$: observation/source positions [m]; $\mathbf{R}_i$: source-to-observer displacement; q_i : source charges [C]; q_0 : test charge; m : mass [kg]; $\mathbf{F}$: force [N]; U : energy [J].

Add signed x/y components first. Like charges repel, unlike attract; gravity attracts. Positive charge follows $\mathbf{E}$, negative charge feels opposite force. Point charges or spherically symmetric exterior fields only.

3 · Continuous charge & trig substitution

$$3.1 \quad Q = \int dq, \quad dq = \lambda_q dl = \sigma dA = \rho_q dV, \quad \mathbf{E} = k_e \int \frac{\mathbf{R}}{R^2} dq$$

$$3.2 \quad E_y = k_e \lambda_q a \int_{-L}^L \frac{dx}{(a^2 + x^2)^{3/2}} = \frac{2k_e \lambda_q L}{a \sqrt{a^2 + L^2}}$$

$$3.3 \quad x = a \tan \theta, \quad dx = a \sec^2 \theta d\theta, \quad R = a \sec \theta$$

$$3.4 \quad E_y = \frac{k_e \lambda_q}{a} \int_{-\theta_0}^{\theta_0} \cos \theta d\theta, \quad \theta_0 = \tan^{-1}(L/a)$$

λ_q : line density [C/m]; σ : surface density [C/m^2]; ρ_q : volume density [C/m^3]; L : source length; A : area; V here: volume. Rod spans $-L \dots L$; observer $(0, a)$, $a > 0$.

Cancel odd components by symmetry. Rod: $L \gg a$ gives $E = 2k_e \lambda_q / a$; $a \gg L$ gives point-charge field with $Q = 2L\lambda_q$. Use $dV = 4\pi r^2 dr$ for spherical density.

4 · Flux, Gauss law & standard geometries

$$4.1 \quad \Phi_E = \int_S \mathbf{E} \cdot d\mathbf{A}, \quad \Phi_E = EA \cos \theta \text{ (uniform)}, \quad \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$4.2 \quad E_{\text{line}} = \frac{\lambda_q}{2\pi\epsilon_0 r}, \quad E_{\text{sheet}} = \frac{\sigma}{2\epsilon_0}, \quad E_{\text{sphere}} = \frac{Q_{\text{enc}}(r)}{4\pi\epsilon_0 r^2}$$

$$4.3 \quad Q_{\text{enc}} = 4\pi \int_0^r \rho_q(s) s^2 ds; \quad \rho_q = \text{const} : E_{\text{in}} = \frac{\rho_q r}{3\epsilon_0}, \quad E_{\text{out}} = \frac{\rho_q R^3}{3\epsilon_0 r^2}$$

$$4.4 \quad E_{\text{cyl, in}} = \frac{\rho_q r}{2\epsilon_0}, \quad E_{\text{cyl, out}} = \frac{\rho_q R^2}{2\epsilon_0 r}$$

Φ_E : flux [$\text{N m}^2/\text{C}$]; θ : angle between $\mathbf{E}$ and outward area normal; Q_{enc} : signed enclosed charge; R : solid-sphere radius; r : distance from center/axis; s : integration radius.

Line: infinite, cylindrical area $2\pi rL$. Sheet: infinite nonconductor, two pillbox faces $2A$. Sphere: $4\pi r^2$; uniform shell has $E = 0$ inside. External charges affect local $\mathbf{E}$ but not net flux. Cylinder: infinitely long, uniform solid, radius R . For concentric shells sum signed enclosed charges by region.

5 · Conductors, potential & work

$$5.1 \quad E_{\text{metal}} = 0, \quad E_t = 0, \quad E_{n, \text{out}} = \sigma / \epsilon_0$$

$$5.2 \quad V = k_e \sum_i \frac{q_i}{R_i} = k_e \int \frac{dq}{R}, \quad \Delta V = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

$$5.3 \quad \mathbf{E} = -\nabla V, \quad E_x = -\frac{\partial V}{\partial x}, \quad U = qV, \quad U_{12} = \frac{k_e q_1 q_2}{r}$$

$$5.4 \quad W_E = -\Delta U = -q\Delta V, \quad K = \frac{1}{2}mv^2, \quad \Delta K = -q\Delta V$$

V : electric potential [$\text{volt} = \text{J/C}$]; U : potential energy; t_n : tangential/normal components; $d\mathbf{l}$: path displacement. K : kinetic energy; W_E : work done by electric field.

Last relation: electric force only. Equipotential $\Rightarrow$ no electric work, $\mathbf{E}$ normal. Conducting cavity with $+q$ induces $-q$ on inner wall. Sphere shell V inside = $k_e Q/R$; V outside = $k_e Q/r$ (zero at infinity).

6 · Useful axial fields & potential shapes

$$6.1 \quad E_{\text{ring}}(z) = \frac{k_e Q z}{(z^2 + R^2)^{3/2}}, \quad V_{\text{ring}}(z) = \frac{k_e Q}{\sqrt{z^2 + R^2}}$$

$$6.2 \quad E_{\text{disk}}(z) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right) \quad (z > 0)$$

$$6.3 \quad F_r = -\frac{dU}{dr}; \quad U = \frac{C_{12}}{r^{12}} - \frac{C_6}{r^6} \Rightarrow F_r = \frac{12C_{12}}{r^{13}} - \frac{6C_6}{r^7}$$

$$6.4 \quad E_{\text{arc, center}} = \frac{2k_e \lambda_q \sin \alpha}{R}, \quad V_{\text{solid, in}} = \frac{k_e Q}{2R} \left(3 - \frac{r^2}{R^2} \right)$$

Uniform ring/disk: R radius, z axial distance. C_{12}, C_6 : positive interaction constants [J m^{12} , J m^6]. Equilibrium: $U' = 0$; stable minimum: $U'' > 0$. Lennard-Jones $r_{\text{eq}} = (2C_{12}/C_6)^{1/6}$.

Positive radial force is outward. Like-charge $U > 0$, opposite-charge $U < 0$ with infinity as zero. A negative potential alone does not determine force: slope does. Arc: total angle 2α , radius R , uniform λ_q ; magnitude along bisector (away from positive arc). Solid sphere: uniform Q , $r < R$.

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7 · Capacitors & dielectric response

$$\begin{aligned} 7.1 \quad C &= \frac{Q}{\Delta V}, \quad C_{\text{plate}} = \frac{\epsilon A}{d}, \quad C_{\parallel} = \sum C_i, \quad \frac{1}{C_{\text{series}}} = \sum \frac{1}{C_i} \\ 7.2 \quad C_{\text{sphere}} &= 4\pi\epsilon \frac{ab}{b-a}, \quad C_{\text{coax}} = \frac{2\pi\epsilon L}{\ln(b/a)} \\ 7.3 \quad U_C &= \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C} = \frac{1}{2} C (\Delta V)^2, \quad u_E = \frac{1}{2} \epsilon E^2 \\ 7.4 \quad \mathbf{D} &= \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon \mathbf{E}, \quad \mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \quad \epsilon = \kappa \epsilon_0 = \epsilon_0 (1 + \chi_e) \end{aligned}$$

C: capacitance [F]; Q: plate-charge magnitude; d: gap; a,b: inner/outer radii; L: coax length; uE: energy/volume [J/m³]; P: polarization, D: displacement [C/m²]; κ , χ_e : dimensionless.

Linear isotropic dielectric fills gap; neglect edges. Parallel shares ΔV ; series shares Q. Battery attached: ΔV fixed, Q rises with κ . Isolated: Q fixed, ΔV and U fall by κ . $\oint \mathbf{D} \cdot d\mathbf{A} = Q_{\text{free}}$.

8 · Dipoles, mechanics & calculus starts

$$\begin{aligned} 8.1 \quad \mathbf{p} &= q\mathbf{d}, \quad \boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}, \quad U = -\mathbf{p} \cdot \mathbf{E} \\ 8.2 \quad E_{\text{axial}} &\simeq 2k_e p/r^3, \quad E_{\text{equator}} \simeq k_e p/r^3, \quad V \simeq k_e p \cos \theta / r^2 \\ 8.3 \quad \sum \mathbf{F} &= m\mathbf{a}, \quad a = \frac{dv}{dt}, \quad v = \frac{dx}{dt} \\ 8.4 \quad a &= \text{const} : v = v_i + at, \quad x = x_i + v_i t + \frac{1}{2} at^2, \quad v^2 = v_i^2 + 2a\Delta x \\ 8.5 \quad a_y &= qE/m, \quad t = \ell/v_x, \quad y = \frac{1}{2} a_y t^2 \quad (v_{y,i} = 0) \end{aligned}$$

p: dipole moment [C m], d points - to +; τ : torque [N m]; θ between p and E (or r for V); far fields require $r \gg d$. ℓ : deflection-plate length, v_x constant. v: particle speed [m/s]; a: acceleration [m/s²]; i: initial value; Δ : final minus initial; m_e : electron mass.

Uniform E: dipole net force zero, torque aligns p; equatorial field opposite p. Integrate a dt with initial v, then v dt with initial x. For Bohr force balance only: $k_e e^2/r^2 = m_e v^2/r$ (not quantization).

9 · Oscillations, energy & damping

$$\begin{aligned} 9.1 \quad F &= -k_s x \Rightarrow \ddot{x} + \omega_0^2 x = 0, \quad \omega_0 = \sqrt{k_s/m}, \quad T = 2\pi/\omega_0 \\ 9.2 \quad x &= A \cos(\omega_0 t + \phi), \quad v = -A\omega_0 \sin(\omega_0 t + \phi), \quad a = -\omega_0^2 x \\ 9.3 \quad U_s &= \frac{1}{2} k_s x^2, \quad K = \frac{1}{2} k_s (A^2 - x^2), \quad E_{\text{tot}} = \frac{1}{2} k_s A^2 \\ 9.4 \quad T_{\text{pend}} &= 2\pi\sqrt{\ell/g}, \quad m\ddot{x} + b\dot{x} + k_s x = F_0 \cos(\Omega t) \\ 9.5 \quad \omega_d &= \sqrt{\omega_0^2 - (b/2m)^2}, \quad A_{\text{drive}} = \frac{F_0}{\sqrt{(k_s - m\Omega^2)^2 + (b\Omega)^2}} \end{aligned}$$

ks: stiffness [N/m]; A: amplitude [m]; ω : angular frequency [rad/s]; ϕ : initial phase; T: period [s]; b: damping [kg/s]; Ω, F_0 : drive frequency/force; $g \approx 9.81 \text{ m/s}^2$.

Ideal SHM: $v_{\text{max}} = A\omega_0$, $a_{\text{max}} = A\omega_0^2$; $x = 0 \Rightarrow$ max speed, $\pm A \Rightarrow v = 0$. Pendulum small angles only. Underdamped amplitude decays as $\exp(-bt/2m)$; damping lowers resonance peak.

10 · Traveling wave: decode & differentiate

$$\begin{aligned} 10.1 \quad y &= A \cos \psi, \quad \psi = k_w x - \omega t + \phi, \quad k_w = \frac{2\pi}{\lambda}, \quad \omega = 2\pi f \\ 10.2 \quad T &= \frac{1}{f} = \frac{2\pi}{\omega}, \quad v_w = \frac{\omega}{k_w} = f\lambda = \sqrt{T/\mu} \\ 10.3 \quad v_y &= \partial_t y = A\omega \sin \psi, \quad a_y = \partial_t^2 y = -\omega^2 y \\ 10.4 \quad \partial_x^2 y &= -k_w^2 y \Rightarrow \boxed{\partial_t^2 y = v_w^2 \partial_x^2 y} \\ 10.5 \quad \mathcal{T} \partial_x^2 y dx &= \mu dx \partial_t^2 y, \quad \bar{P} = \frac{1}{2} \mu \omega^2 A^2 v_w \end{aligned}$$

y: transverse displacement; x: position; k_w : wave number [rad/m]; λ : wavelength [m]; f: frequency [Hz]; μ : string mass/length [kg/m]; T: tension [N]; $\bar{P}$: mean power [W].

Minus ωt travels +x; plus travels -x. Constant phase gives direction. Pattern speed v_w differs from particle velocity v_y . Small slopes and constant tension/density give the linear PDE.

11 · Boundaries, interference & standing waves

$$\begin{aligned} 11.1 \quad \Delta\psi &= k_w \Delta x - \omega \Delta t + \Delta\phi, \quad A_{\text{sum}} = 2A |\cos(\Delta\psi/2)| \\ 11.2 \quad y_1 + y_2 &= 2A \sin(k_w x) \cos(\omega t), \quad f_n = \frac{nv_w}{2L} \quad (n = 1, 2, \dots) \\ 11.3 \quad f_n &= \frac{nv_w}{4L} \quad (n = 1, 3, 5, \dots), \quad f_{\text{beat}} = |f_1 - f_2| \\ 11.4 \quad n_1 \sin \theta_1 &= n_2 \sin \theta_2, \quad n = c/v, \quad \sin \theta_c = n_2/n_1 \\ 11.5 \quad Z_j &= \mathcal{T}/v_j, \quad r_A = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t_A = \frac{2Z_1}{Z_1 + Z_2} \end{aligned}$$

L: vibrating length; n harmonic integer except in refraction, where n is index and c is vacuum light speed. θ measured from normal; total internal reflection needs $n_1 > n_2$.

Fixed string / open-open pipe: all harmonics; open-closed pipe: odd only. Nodes spaced $\lambda/2$. Fixed reflection inverts, free does not. Constructive phase $2\pi m$; destructive $(2m+1)\pi$. Boundary f unchanged; Snell follows $k_1 \sin \theta_1 = k_2 \sin \theta_2$. String junction, equal tension: Z impedance; rA, tA signed displacement-amplitude ratios. Power fractions rA^2 and $(Z_2/Z_1)tA^2$ sum to 1.

12 · Sound, levels & Doppler

$$\begin{aligned} 12.1 \quad v_s &= \sqrt{B/\rho_m}, \quad I = \frac{P}{4\pi r^2} = \frac{p_{\text{rms}}^2}{\rho_m v_s} = \frac{1}{2} \rho_m v_s \omega^2 s_m^2 \\ 12.2 \quad p_m &= \rho_m v_s \omega s_m, \quad p_{\text{rms}} = p_m/\sqrt{2}, \quad \beta = 10 \log_{10}(I/I_0) \\ 12.3 \quad \Delta\beta &= 10 \log_{10}(I_2/I_1) = 20 \log_{10}(p_2/p_1) \\ 12.4 \quad f_o &= f_s \frac{v_s + v_o}{v_s - v_{\text{src}}}, \quad \mathcal{E}_{\text{ind}} = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int \mathbf{B} \cdot d\mathbf{A} \end{aligned}$$

B: bulk modulus [Pa], ρ_m : medium density [kg/m³], s_m : displacement amplitude; p_m, p_{rms} : pressure [Pa]; $I_0 = 10^{-12} \text{ W/m}^2$; air reference $p_0 = 20 \text{ } \mu\text{Pa}$. β : dB. Last two: scan induction reminder; magnetic B [T], Φ_B [Wb], emf [V]. f_o, f_s : observed/source frequency [Hz]; v_s : sound speed; v_o, v_{src} : observer/source velocity [m/s].

Doppler velocities relative to medium: v_o positive toward source, v_{src} positive toward observer (subsonic). Approach raises f, shortens observed period. Pressure $\times 3 \Rightarrow$ intensity $\times 9 \Rightarrow +9.54 \text{ dB}$. Pitch: f; timbre: spectrum; sound needs medium.